

Red Panda Optimization Algorithm-Based PID Controller Design for Automobile Cruise Control System

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Abstract—The integration of cruise control into vehicles plays a crucial role in the domain of advanced driver assistance systems. Functionally, cruise control relies on a Proportional-Integral-Derivative (PID) controller to maintain the vehicle's velocity at a predetermined setpoint. Traditional parameter tuning of the PID controller often results in oscillatory output, prolonged rise time, and extended settling duration. To address these issues, we introduce the Red Panda Optimization (RPO) algorithm to optimize the PID controller parameters, thereby enhancing both the accuracy and stability of the vehicle's cruise control mechanism. By accounting for the inherent nonlinearities and dynamic characteristics of the system, the RPO algorithm determines the controller gain while simultaneously identifying the optimal value through an evaluation of the integral time square error. To validate and verify the performance of our proposed system, we conducted simulations using a Bode plot within the MATLAB R2023a framework. The implemented algorithm for the adaptive cruise control system demonstrates improved responsiveness and heightened system stability, as evidenced by reduced rise time, minimized latency, and diminished overshoot.

Keywords—ACC System, Optimization Algorithm, PID Controller Design, Red Panda Optimization Algorithm

I. INTRODUCTION

The combination of technological innovation and increasing interest in emerging vehicles has led to a radical transformation in the automotive sector over the past decade. Adaptive Cruise Control (ACC) is a sophisticated driver-assistance system tailored for road vehicles, operating autonomously to manage the vehicle's speed. The Society of Automotive Engineers has categorized driving automation into levels 0 to 6, with the integration of Adaptive Cruise Control (ACC) starting from level 1 and onward [1], [2]. In 1992, Mitsubishi Motors implemented the ACC system, which detected neighboring objects and provided warnings to the driver. By 1995, Mitsubishi Motors had incorporated ACC to regulate speed. Many manufacturers, such as those producing the Diamante sedan, Distronic, and Debonair, adopted the ACC system, with regulation provided by the European standard ISO 15622 [3]. ACC is now widely used worldwide, increasing the usage of self-driving vehicles. According to a 2020 survey by the U.S. Department of

Transportation, around 74% of passenger vehicles on U.S. roads were equipped with cruise control [4], [5]. This technology enhances passenger safety, maintains the optimum distance between vehicles, and minimizes driver errors, with the primary objective being to reduce manual intervention while maximizing different aspects of vehicle performance [6]. Recent vehicle models such as the Hyundai Elantra, Hyundai Sonata, Mahindra XUV700, MG Gloster, Toyota Camry, Volkswagen Golf GTI, Volvo V60 Recharge, BMW 5 Series, and Audi A6 have incorporated features of ACC systems [7]. India, one of the fastest-growing countries in the automobile sector, boasts 205 different car manufacturers throughout the country offering cruise control. Currently, popular cruise control-equipped cars include the Hyundai i2, Hyundai i20 N Line, Mercedes-Benz EQE SUV, and Tata Nexon. India is poised to claim the third spot globally by 2030 in the automotive industry [8].

The Adaptive Cruise Control (ACC) system facilitates effective transportation by offering various advantages, including increased safety, reduced driver fatigue, enhanced convenience, improved fuel efficiency, efficient traffic control, and collision avoidance [9]. Cruise control, found in many modern vehicles, enables drivers to keep a steady speed without constantly pressing the gas pedal. This feature boosts fuel efficiency on highways by 4-10% and reduces driver tiredness during long journeys. Additionally, it contributes to improved traffic flow. Developing new approaches with better effectiveness is crucial for achieving optimal control of the ACC system [10]. Accurately determining the system parameters of the ACC system relies on analyzing real-time driving data, including variations from the predetermined speed and adjustments made to accommodate different driving conditions in the past, present, and anticipated future. As the ACC system is nonlinear, the controller is designed to ensure steady operation. A proportional-integral-derivative (PID) controller is commonly utilized, with its gain determined using traditional methods such as Ziegler-Nichols and Cohen-Coon [11]. The system exhibits an oscillatory response, longer rise time, and settling time, necessitating improvement in the response, achievable through proper tuning of the PID controller. Finding an appropriate gain for the ACC system through control algorithms ensures better

performance. Metaheuristic algorithms and soft computing techniques are often employed for PID controllers [12]. In this study, the RPO algorithm is suggested for adjusting the PID controller parameters to improve the precision and reliability of the vehicle's cruise control system. The content is organized as follows: Section II discusses related works, Section III presents the ACC system model and transfer function, Section IV elaborates on PID control commonly used in ACC systems, Section V discusses metaheuristic algorithms for ACC systems and proposes the RPO algorithm for finding controller parameters, and Section VI provides the conclusion of the proposed work.

II. RELATED WORKS

The PID controller is a widely used control loop feedback mechanism employed in various engineering applications [13]. In ACC systems, the PID controller assists in modifying speeds by incorporating the proportional, integral, and derivative elements of the variance between the desired speed or distance and the actual speed or distance concerning the vehicle ahead [14]. The proportional term enables the system to respond proportionally to the current error, the integral term aids in eliminating any steady-state error, and the derivative term assists in predicting the system's future behavior based on its current rate of change [15]. By incorporating PID control in ACC systems, vehicles can smoothly and stably adjust their speed, ensuring a safe distance from the vehicle ahead while maintaining a consistent cruising speed. This integration of PID control in ACC contributes to enhanced driving comfort, increased safety, and improved traffic flow on highways and roads [16], [17]. The PI or PID controller is typically used for ACC systems, and the tuning parameters of PID controllers are determined using the Ziegler-Nichols method. However, complex systems may take longer to reach a stable state. Metaheuristic methodologies have been utilized to quickly adjust the controller parameters [18]. Various optimization techniques, such as the Arithmetic Optimization Algorithm (AOA), Cuckoo Search Optimization, Differential Evolutionary Algorithm, Simulated Annealing Algorithm (SAA), Harris Hawks Optimization (HHO), Improved Kidney-Inspired Algorithm (IKA), Firefly Algorithm, Genetic Algorithm, and Opposition-based AOA with the Nelder-Mead (ObAOANM) search method [1], as well as the Particle Swarm Optimization Algorithm, have been applied in ACC systems [19], [20]. Ant Lion Optimizer (ALO) to optimize the tuning parameters for a PID-controlled cruise control system, aiming to regulate performance indices for automobile cruise control [21]. A PID controller using dominant pole placement to stabilize an initially unstable cruise control model, and optimized the controller parameters using Genetic Algorithm (GA) [22]. These algorithms often incorporate principles such as self-organization, decentralized control, and information sharing, enabling vehicles to collectively respond to fluctuations in traffic patterns and avoid congestion or collisions. By applying bio-inspired algorithms, ACC systems can enhance their adaptability and responsiveness, leading to improved traffic management, reduced fuel consumption, and heightened safety on the roads. This approach leverages the efficiency and resilience observed in natural systems, thereby contributing to the development of intelligent and sustainable transportation solutions [23]. Therefore, in this research, the RPO algorithm is utilized to optimize the PID controller parameters to enhance the reliability, adaptability, and responsiveness of emerging automotive vehicles. The

performance of the controllers is evaluated using the objective function of Integral Time Square Error (ITSE).

III. ACC SYSTEM MODEL

In the ACC system, the vehicle's velocity is altered according to the driver's selected speed preference. The variance between the set speed and the actual speed is termed as an error, and adjustments to the engine throttle angle are made to manage this error. This operation involves adjusting the driving force F_d , and the vehicle's longitudinal dynamics are described in (1).

$$F_d = M \frac{dv}{dt} + F_a + F_g \quad (1)$$

In this context, F_d stands for aerodynamic drag, while F_g represents the downgrade force. $M \frac{dv}{dt}$ gives the inertia force. Fig. 1 shows the dynamic model of the ACC system.

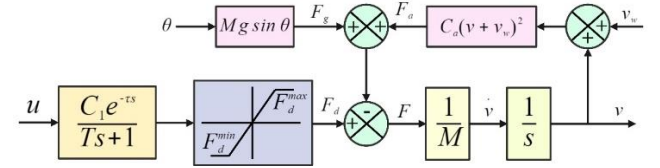


Fig. 1. Dynamic model of ACC system

The system model demonstrates characteristics of a first-order lag system with a saturation feature. The details of the system model's parameters can be found in Table I. The vehicle's speed is assumed to be 30 km/h without slope resistance or the influence of wind gusts. The assumptions of the system model are well-described in [1], and the model is presented in (2) and (3).

$$v = \frac{1}{M} (F_d - C_a v^2) \quad (2)$$

$$F_d = \frac{1}{T} (C_1 u(t - \tau) - F_d) \quad (3)$$

For 30 kmph, the equilibrium point requires the nominal drive force of $F_{d0} = C_a v_0^2$ and nominal throttle position of $u_0 = F_{d0}/C_1$. Equations (4) and (5) gives the linearizing model of the system

$$\delta v = -\frac{2C_a v_0}{M} \delta v + \frac{1}{M} \delta F_d \quad (4)$$

$$\delta F_d = -\frac{1}{T} \delta F_d + \frac{C_1}{T} \delta u(t - \tau) \quad (5)$$

The linearized model is simplified further [1] and substituted the Table I values. The obtained transfer function is given in (6).

$$G_{sys}(s) = \frac{\Delta V(s)}{\Delta U(s)} = \frac{2.4767}{(s+0.0476)(s+1)(s+5)} \quad (6)$$

TABLE I. ACC SYSTEM MODEL PARAMETERS

Parameter	Terms	Value	Unit in SI
C_1	Actuator constant	742	-
C_a	Aerodynamic drag coefficient	1.18	N/(m/s) ²
g	Gravitational acceleration	9.8	m/s ²
M	Mass of vehicle and passenger	1500	Kg
T	Observation time	1	s
F_d^{min}	Minimum saturation limit	-3500	N
F_d^{max}	Maximum saturation limit	3500	N
τ	Driver reaction time	0.2	s

The ACC system is integrated with a PID controller, where the PID controller transfer function can be found in (7). The controller has three parameters, namely k_p , k_i and k_d . Equation (8) provides the system's complete transfer function. The range of the gain is specified as $k_p^{min} < k_p < k_p^{max}$, $k_i^{min} < k_i < k_i^{max}$, $k_d^{min} < k_d < k_d^{max}$ and its values are $3 < k_p < 5$, $0.10 < k_i < 0.25$, $3 < k_d < 5$.

$$G_{pid}(s) = \frac{k_d s^2 + k_p s + k_i}{s} \quad (7)$$

$$G_{CLTF}(s) = \frac{G_{sys}(s)G_{pid}(s)}{1 + G_{sys}(s)G_{pid}(s)} = \frac{2.4767(k_d s^2 + k_p s + k_i)}{s(s + 0.0476)(s + 1)(s + 5) + 2.4767(k_d s^2 + k_p s + k_i)} \quad (8)$$

The controller's gain within the specified range is determined in the RPO algorithm, and the optimal value is obtained by assessing the ITSE as defined in (9). The PID controller system implemented with RPO is depicted in Fig. 2.

$$ITSE = \int_0^T t [e(t)]^2 dt \quad (9)$$

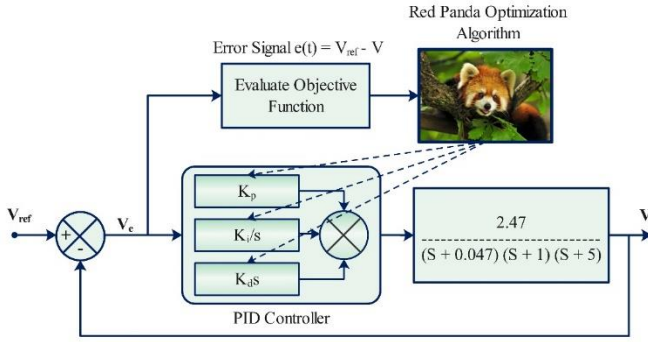


Fig. 2. System with RPO implemented PID Controller

IV. RED PANDA OPTIMIZATION ALGORITHM

The Red Panda is an herbivorous mammal, and its inspiration was discussed in reference [22]. This animal lives in temperatures ranging between 17°C and 25°C and is mostly endemic to the Himalayas, Laos, Nepal, and Southern China. Red Pandas search for food quickly and with agility. As an essential part of their feeding process, they use their front paws to place food into their mouths. These animals primarily consume bamboo but also eat acorns, berries, fruits, grasses, lichen, mushrooms, and roots. Pandas are solitary animals, and their reproductive system involves both male and female pandas. After a gestation period, the female panda gives birth, and the cubs stay with their mother. Red Pandas face threats from predators such as snow leopards, martens, and humans. Their typical activities involve foraging strategies that rely on their hearing, sight, smell, and tree-climbing abilities. Due to the natural behaviors of these organisms, mathematical modeling was developed, resulting in RPO, a population-based metaheuristic algorithm. Each Red Panda represents an option for responding to the problem. The scope of the search is determined by the number and values of the problem variables. Equation (10) is utilized to generate an initial population randomly.

$$x_{ij} = lb_j + rand().(ub_j - lb_j); \quad i = 1, 2, \dots, N; \quad j = 1, 2, \dots, m \quad (10)$$

Here lb and ub is the lower bound and upper bound of problem variables respectively. The $rand()$ function produces

a random number between 0 and 1, indicating the quantity of Pandas and serving as the number of problem variables. Finally, the initial population is formed in matrix form as in (11).

$$X = \begin{bmatrix} X_1 \\ \vdots \\ X_i \\ \vdots \\ X_N \end{bmatrix}_{N \times m} = \begin{bmatrix} x_{11} & \cdots & x_{1j} & \cdots & x_{1m} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{i1} & \cdots & x_{ij} & \cdots & x_{im} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{N1} & \cdots & x_{Nj} & \cdots & x_{Nm} \end{bmatrix}_{N \times m} \quad (11)$$

The objective function is evaluated for each candidate solutions and represented in (12).

$$OF = \begin{bmatrix} OF_1 \\ \vdots \\ OF_i \\ \vdots \\ OF_N \end{bmatrix}_{N \times 1} = \begin{bmatrix} OF(X_1) \\ \vdots \\ OF(X_i) \\ \vdots \\ OF(X_N) \end{bmatrix}_{N \times 1} \quad (12)$$

Here OF_i indicates the objective function of i^{th} panda. OF indicates the objective function values of initial population of vector size $N \times 1$. The objective function is the primary focus for finding the best solutions, and it is updated in each iteration of the exploration and exploitation phases.

A. Exploration Phase

The Red Pandas excel in locating food sources due to their exceptional senses of smell, hearing, and vision. Each Red Panda heads towards the food source, which is considered the optimal objective function. Each Red Panda's allocation to a food source is determined by evaluating the values of the objective function described in (13). One of the potential locations is randomly selected to serve as the designated food position.

$$FS = \{X_k | k \in \{1, 2, \dots, N\} \text{ and } F_k < F_i\} \cup \{X_{best}\} \quad (13)$$

In this context, FS represents the expected food sources for the i^{th} Red Panda, while X_{best} indicates the location of the The Red Panda that yields the greatest value for the objective function is the one that moves towards the food source, leading to notable shifts in their positions. The foraging behavior of Red Pandas is depicted in (14) and (15), which model their movements during this stage.

$$x_{i,j}^{new,fp} = x_{i,j} + rand().(sFS_{ij} - l.x_{i,j}) \quad (14)$$

$$X_i = \begin{cases} x_{i,j}^{new,fp}, & F_i^{new,fp} < F_i; \\ X_i, & \text{otherwise} \end{cases} \quad (15)$$

In the text, $x_{i,j}^{new,fp}$ represents the updated position of the i^{th} Red Panda, while sFS_{ij} denotes the chosen food sources at the j^{th} location. The value of l is a randomly chosen number from the set $\{1, 2\}$.

B. Exploitation Phase

The Red Pandas take breaks in the trees and move around slightly after searching for food on the ground, resulting in small changes in their positions. This behavior is mathematically modeled in (16) and (17).

$$x_{ij}^{p2} = x_{ij} + \frac{lb_j + rand().(ub_j - lb_j)}{t}; \quad i = 1, 2, \dots, m; \quad j = 1, 2, \dots, N; \quad t = 1, 2, \dots, T \quad (16)$$

$$X_i = \begin{cases} x_{ij}^{p2}, & F_i^{p2} < F_i \\ X_i, & \text{otherwise} \end{cases} \quad (17)$$

where X_i^{p2} is the new position of the i^{th} Red Panda and x_{ij}^{p2} is its j^{th} dimension, F_i^{p2} gives the objective function, the current iteration number is represented by t , while T stands for the maximum number of iterations.

V. RESULT AND DISCUSSION

The ACC system without a controller exhibits three poles, resulting in a smooth response but with a high amplitude of the set value. The system model with the controller is depicted in Fig. 2. The RPO algorithm identified the controller's gain. Initially, a population of 30 was randomly generated within specified ranges and underwent two phases to determine the optimal controller gain. These two phases are executed in each iteration of the RPO algorithm, and the best controller gain is recorded during the exploitation phase.

The RPO algorithm was executed for 50 iterations, using an ITSE objective function to evaluate the best solutions. The Table II presents the optimal values for the controller gain. The performance of the proposed RPO algorithm yielded controller gains of 4.3017 for proportional, 0.1117 for integral, and 4.08064 for derivative components, in comparison to the gains obtained using AOA and ObAOANM algorithms. As a result of the controller's amplification, the system now showcases four poles and two zeros located on the complex plane left side. These poles demonstrate a preference for the characteristics of the ACC system. Specifically, two poles are complex, while the remaining poles are real zeros. Complex poles enable control over the frequency response of a system. By adjusting the real and imaginary parts of the poles, the system's response can be shaped to meet desired specifications, such as bandwidth and damping. Furthermore, the complex conjugate nature of complex poles promotes smoother responses.

TABLE II. GAIN OF THE PID CONTROLLER

Algorithms	k_p	k_i	k_d
AOA [1]	4.0407	0.2119	4.1548
ObAOANM [1]	4.1458	0.1875	4.3965
ALO [21]	3.7284	0.17362	3.5161
GA [22]	3.5907	0.163	3.3021
RPO [proposed]	4.3017	0.1117	4.8064

The velocity step response of the ACC system is shown in Fig. 3, and the Bode response is depicted in Fig. 4. The PID controller system, utilizing the suggested RPO algorithm, demonstrated enhanced performance. The rise time (T_r), settling time (T_s), peak time (T_p) was decreased in the proposed controller gain utilization in the ACC system. The gain of k_p , k_d was high as compared with [1]. The increased gain of k_p decreases the T_r , steady state error (E_{ss}) and the increased gain of k_d decreased the T_r , T_s , Overshoot (M_p). The measured time domain specifications are given in Table 3. The gain of k_i was less value also controls the T_r , T_s , T_p , M_p and E_{ss} . The stability of the system was analyzed through frequency response, and a stable state was observed. Table III indicates the bandwidth of the proposed controller, which is higher compared to [24]. Based on the results obtained from the proposed controller, the time domain response of the system exhibits improvements in several parameters. Specifically, the mean values of maximum peak-overshoot, steady state error, rise time, settling time, and peak-overshoot are 12.06%, 0, 0.6921, 1.0808, and 1.3919 respectively, compared to the values reported in [1], [21], [22].

This improvement enhances the control loop performance, reduces delay and overshoot, and ensures better stability, precision, and robustness.

The RPO algorithm was run 40 times, and the optimal gain value was identified. The objective function is to choose the optimal solution. Table IV presents the statistical analysis of the ITSE objective function, indicating that the RPO algorithm produces lower mean, standard deviation, best, and worst values. The proposed PID controller based on the RPO algorithm achieved an ITSE error value of 0.0704 compared to [1].

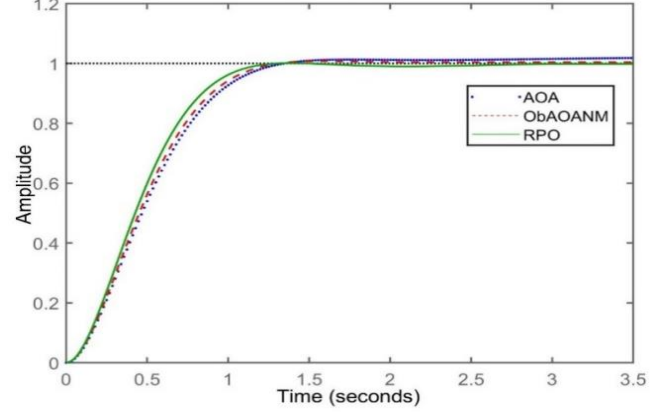


Fig. 3. Step response of ACC system

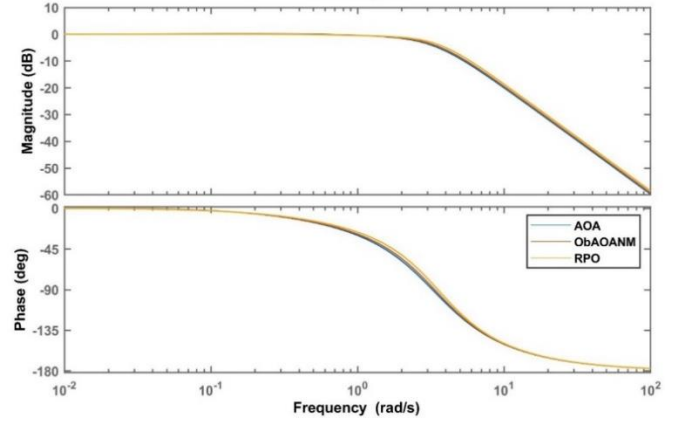


Fig. 4. Bode plot response of ACC system

TABLE III. TIME DOMAIN SIMULATION RESULTS

Controller	$\%M_p$	$\%E_{ss}$	$T_r(s)$	$T_s(s)$	$T_p(s)$
AOA [1]	0.3625	0	0.7797	1.2201	1.6369
ObAOANM [1]	0.0108	0	0.7457	1.1729	1.5188
ALO [21]	0.9442	0	0.8895	1.3828	2.2001
GA [22]	1.1400	0	0.9450	1.4600	2.1500
RPO [proposed]	0.1206	0	0.6921	1.0808	1.3919

TABLE IV. STATISTICAL ANALYSIS OF ITSE OBJECTIVE FUNCTION

Statistical Measure	AOA	ObAOANM	RPO
Mean	0.085056	0.077181	0.070410
Standard Division	0.002666	0.001244	0.000123
Best	0.082147	0.075680	0.070314
Worst	0.090608	0.079918	0.071825

VI. CONCLUSION

In this study, we suggest the use of the RPO algorithm to improve both the efficiency and stability of the vehicle's cruise control system through the optimization of the PID controller's parameters. Through the exploitation phase, the RPO

algorithm effectively identifies the optimal controller gain by evaluating the ITSE objective function. To validate our approach, we conducted simulations in the MATLAB R2023a environment, utilizing Bode plots for analysis. The system's stability was evaluated through both frequency and time responses. Notably, improvements were observed in the time-domain response parameters and bandwidth. Our approach yielded a significantly lower ITSE error value of 0.0704 compared to alternative algorithms such as AOA and ObAOANM. Moreover, our simulated results indicate that our proposed algorithm exhibits faster responsiveness and enhanced system stability in the ACC system.

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